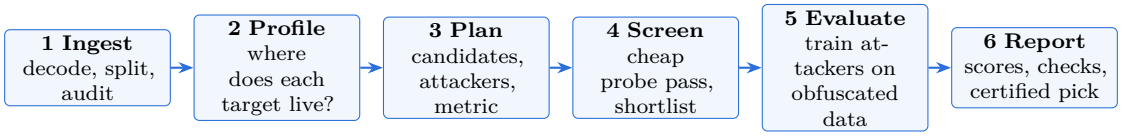
privharness: how the privacy harness works

`research/privacy-obfuscation/privharness`

Input: any labelled video or image dataset with a *task* to keep (e.g. the action) and one or more *secrets* to hide (e.g. who is in it, appearance attributes). Output: the obfuscator with the best measured privacy–utility trade-off, certified by attackers trained on its output.



1. Ingest

(`ingest.py`, `adapters/`)

- A JSON description names the media (table or folder + regex), which columns are task, secrets and groups (recording, scene, session), and how to split. Media are decoded to uint8 frame arrays at a common size.
- Splits never put one recording on both sides. The audit flags secret classes missing from a side, tiny identity sets, and **near-duplicate footage across the split** (appearance hash gate, then motion-trace correlation $\geq$ 0.99; 5/5 planted copies found on KTH, no false alarms).

2. Profile

(`profile.py`, `views.py`)

- Each item is turned into *cue views* that keep one kind of information: scene with people blacked out, person crop, body keypoints and proportions, posture only, motion, colour, low resolution, and the task label alone.
- A linear probe per (target, view) gives *predictability*: 1.0 = the view tells as much as raw video. E.g. KTH: action lives in posture (1.0); identity in appearance and scene (0.9).

3. Plan

(`planner.py`, `candidates.py`)

- Feasibility: temporal methods need video; stick figures need detectable skeletons.
- Cue matching: e.g. secret in body proportions, task in posture $\Rightarrow$ canonical stick figures first.
- If the task label alone predicts a secret ($\geq$ 75%), that secret is scored *beyond the task* (conditional metric).
- Catalogue: blur, downsampling, person removal/blur, motion maps, stick figures, canonical stick figures ($\times$1–32), and learned BDQ / BDQ$\rightarrow$pool trained against *all* secrets at once.

4. Screen

(`screen.py`)

- Every fixed candidate obfuscates a sample; frozen-feature probes attack it. The Pareto front and best proxy scores form the shortlist. Agreement with full evaluation is reported (rank correlation 0.76–1.00 so far).

5. Evaluate

(`evaluate.py`, `attackers.py`, `protocols.py`)

- The whole dataset is obfuscated (8-bit, as released). Fresh task models and fresh **attackers are trained on the obfuscated training set** and tested on obfuscated test data: frame CNN, video CNN, skeleton/gait network; frozen raw-trained attackers are also run (they understate leakage by a median 17–82 points).
- Protocols: same split; unseen actions; unseen action + unseen scene; unseen group values.

6. Scores

(`scoring.py`, `report.py`)

$$\text{baseline} = \max(\text{chance},\ \text{task-label-only attacker})$$

$$\text{leakage} = \frac{A_{\text{obf}} - \text{baseline}}{A_{\text{raw}} - \text{baseline}}, \qquad \text{privacy} = 1 - \text{clip}(\text{leakage})$$

$$\text{utility} = \frac{u_{\text{obf}} - \text{chance}_u}{u_{\text{raw}} - \text{chance}_u}, \qquad \text{PUS} = \frac{2\,P\,U}{P+U}$$

- A = the **best** attacker (worst case). 95% intervals by bootstrapping test recordings.
- **Certified** privacy = worst over the pick and its re-runs (2 more seeds, 2$\times$ training, DINOv2 attackers).
- Checks: raw footage measurably above baseline; all-black video must score at baseline.

Validation (one search per dataset, no hand tuning)

dataset	harness pick	utility	privacy*
KTH (25 people)	canon. stick fig. $\times$16	88.8	89.5
KTH, new scenario	canon. stick fig. $\times$16	75.5	84.8
VP-HMDB51 (attributes)	blur people only	63.3	43.4
UTD-MHAD (8 people)	1$\times$2 px motion map	53.7	51.9
Charades (90 homes)	running		

*certified (worst over re-runs), 0–100.

- On KTH it independently picked the method found by hand; on UTD-MHAD it overturned a hand-drawn conclusion; on VP-HMDB51 its robustness pass caught a stronger attacker.

Run it

```
python -m privharness ingest desc.json ROOT
python -m privharness search ROOT
```

Design and references: `docs/harness.md`. Next: a sealed test set for selection; an LLM that proposes candidates and attacks, judged by the harness.